

Standard Operating Procedure

Shimadzu SSM-5000A TOC Analyzer
for Soil Samples with Varying Carbonate Concentrations

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1 Purpose and Scope

This SOP describes the procedure for determining Total Carbon (TC), Total Inorganic Carbon (TIC), and Total Organic Carbon (TOC) in soil samples using the Shimadzu SSM-5000A TOC Analyzer equipped with a Solid Sample Module. This procedure is applicable to soil samples with high ($> 10\%$), medium ($2\text{--}10\%$), and low ($< 2\%$) carbonate concentrations.

2 Principle

The Shimadzu SSM-5000A system uses high-temperature combustion (900°C) to convert all carbon in the sample to CO_2 , which is then quantified by NDIR (Non-Dispersive Infrared) detection. For carbonate-bearing soils:

- **TC (Total Carbon)** = measured by direct combustion
- **TIC (Total Inorganic Carbon)** = measured by acid treatment (200°C)
- **TOC (Total Organic Carbon)** = $\text{TC} - \text{TIC}$

3 Safety and Precautions

3.1 Personal Protective Equipment (PPE)

- Safety glasses or goggles
- Laboratory coat
- Heat-resistant gloves when handling hot ceramic boats
- Nitrile gloves when handling acids

3.2 Chemical Hazards

- Phosphoric acid (25% solution): corrosive, causes severe burns
- High-purity oxygen: oxidizer, keep away from combustibles
- Soil samples: may contain pathogens or heavy metals

3.3 Equipment Hazards

- Combustion furnace reaches 900°C
- Acid furnace reaches 200°C
- Allow adequate cooling time before handling boats

4 Equipment and Materials

4.1 Instrumentation

- Shimadzu SSM-5000A TOC Analyzer with Solid Sample Module
- High-purity oxygen supply (99.999% minimum purity)
- Carrier gas supply (oxygen, 99.999% minimum purity)
- Computer with SSM-5000A Control Software (v3.x or higher)
- Analytical balance (0.0001 g precision, capacity ≥ 220 g)
- Autosampler unit (40-position carousel)

4.2 Consumables

- Ceramic sample boats (platinum or high-purity alumina, pre-combusted)
- 25% phosphoric acid solution (prepared fresh weekly)
- Glucose anhydrous (certified reference material, 99.9% purity)
- Calcium carbonate standard (reagent grade, dried at 105°C for 2 hours)
- Quartz wool for combustion tube packing
- Halogen moisture trap cartridge

4.3 Sample Preparation Equipment

- Mortar and pestle (agate or alumina)
- Ball mill (optional, for fine grinding)
- 2 mm and 0.25 mm sieves (stainless steel or brass)
- Drying oven (105°C $\pm$ 2°C)
- Desiccator with indicating desiccant (drierite or silica gel)

5 Reagents and Standards

5.1 Phosphoric Acid Solution (25% v/v)

1. Add 250 mL concentrated phosphoric acid (85%) slowly to 600 mL deionized water in a 1000 mL volumetric flask
2. **CAUTION:** Add acid to water, never water to acid
3. Mix thoroughly by swirling (exothermic reaction)
4. Allow to cool to room temperature
5. Dilute to 1000 mL with deionized water
6. Store in amber glass bottle with screw cap
7. Label with: name, concentration, preparation date, expiration date (7 days)
8. Prepare fresh weekly or when contamination is suspected

5.2 TOC Calibration Standards (from Glucose)

Stock Solution Preparation (1000 mg C/L):

1. Weigh exactly 2.500 g $\pm$ 0.001 g anhydrous glucose into 1000 mL volumetric flask
2. Glucose contains 40.0% carbon by weight
3. Dissolve in approximately 800 mL deionized water
4. Dilute to volume with deionized water
5. Mix thoroughly by inversion (20 times)
6. Store at 4°C for up to 7 days

Working Standard Preparation:

5.3 TIC Calibration Standards (from Calcium Carbonate)

Preparation:

1. Dry reagent-grade CaCO₃ at 105°C for 2 hours
2. Cool in desiccator for minimum 30 minutes
3. Calculate theoretical carbon content: CaCO₃ molecular weight = 100.09 g/mol; C = 12.01 g/mol
4. Carbon content = $\frac{12.01}{100.09} \times 100 = 12.0\%$ by weight
5. Prepare solid standards by weighing directly into ceramic boats:

Level	Target Conc. (mg C/L)	Stock Vol. (mL)	Final Vol. (mL)	Mass C (mg)
1	10	1.0	100	0.001
2	25	2.5	100	0.0025
3	50	5.0	100	0.005
4	100	10.0	100	0.010
5	250	25.0	100	0.025
6	500	50.0	100	0.050

Table 1: TC Calibration Standard Preparation from Stock Solution

Level	CaCO ₃ Mass (mg)	Carbon Mass (mg C)	Expected Peak Area (est.)	Acid Volume (μL)
1	8.3	1.0	Low	150
2	41.7	5.0	Medium-Low	150
3	83.3	10.0	Medium	200
4	208.3	25.0	Medium-High	200
5	416.7	50.0	High	250

Table 2: TIC Calibration Standard Preparation

- Store prepared standards in desiccator
- Use within 24 hours of preparation
- Weigh standards to nearest 0.1 mg

6 Sample Preparation

6.1 Initial Processing

1. Air-dry soil samples at room temperature (3–7 days) or in drying oven at 40°C maximum (to prevent organic matter loss)
2. Remove large debris, roots, stones, and organic fragments manually
3. Break up aggregates gently using mortar and pestle
4. Pass through 2 mm sieve for initial homogenization
5. Record percent of sample retained on sieve (coarse fraction)

6.2 Fine Grinding

1. Take representative subsample (approximately 50 g) from <2 mm fraction
2. Further grind using agate mortar and pestle or ball mill
3. Grind until all material passes through 0.25 mm sieve (60 mesh)
4. For ball mill: grind for 5–10 minutes at medium speed
5. Mix thoroughly by quartering method or tumbling for 2 minutes
6. Store ground samples in airtight containers (polypropylene or glass)

6.3 Moisture Determination

1. Weigh 2–5 g of ground sample into pre-weighed, pre-dried aluminum crucible
2. Record crucible + wet sample weight to 0.0001 g
3. Dry at 105°C ± 2°C for 24 hours in forced-air oven
4. Cool in desiccator for minimum 30 minutes
5. Reweigh crucible + dry sample to 0.0001 g

6. Calculate moisture content:

$$\text{Moisture \%} = \frac{\text{Wet weight} - \text{Dry weight}}{\text{Wet weight}} \times 100 \quad (1)$$

- Perform moisture determination in triplicate
- Use average moisture content for all calculations
- Redetermine moisture if sample storage exceeds 7 days

6.4 Sample Weighing Guidelines

Soil Type		TC Analysis	TIC Analysis	Expected TIC
High	Carbonate	10–20 mg	5–10 mg	> 1.2% C
(> 10%)				
Medium	Carbonate (2–	20–40 mg	10–30 mg	0.24–1.2% C
10%)				
Low	Carbonate (< 2%)	40–80 mg	30–80 mg	< 0.24% C
Non-calcareous		50–100 mg	N/A	0% C

7 Detailed Instrument Startup Procedure

7.1 Pre-Startup Inspection (Day Before Analysis)

7.1.1 Gas Supply Verification

1. Locate oxygen cylinder and check pressure gauge
2. Verify cylinder pressure > 500 psi (34 bar)
3. If pressure < 500 psi, arrange for cylinder replacement
4. Inspect regulator for damage, corrosion, or leaks
5. Verify gas line connections are tight (hand-tight + 1/4 turn with wrench)
6. Check for leaks using leak detection solution or soap solution:
 - Apply solution to all fittings and connections
 - Open cylinder valve slightly
 - Look for bubble formation indicating leaks
 - Tighten connections if bubbles appear
 - Retest after tightening
7. Verify gas tubing is not kinked, cracked, or discolored

7.1.2 Instrument Physical Inspection

1. Open instrument doors and visually inspect:
 - Combustion tube for cracks or deposits
 - Ceramic boats for contamination or damage
 - Quartz wool packing (replace if discolored or compressed)
 - Acid delivery system for crystallization or leaks
 - NDIR detector cell window for fogging or deposits
2. Check autosampler carousel:

- Verify smooth rotation (rotate manually)
 - Inspect positioning mechanism for alignment
 - Check boat holder clips for wear
3. Verify drain bottles are empty (empty if > 50% full)
 4. Check desiccant color in moisture trap (replace if pink/saturated)

7.2 System Power-Up Sequence (Analysis Day)

7.2.1 Step 1: Turn On Gas Supply

1. Slowly open oxygen cylinder main valve (turn counterclockwise)
2. Turn valve 1.5–2 full rotations (not fully open)
3. Observe high-pressure gauge on cylinder (should read cylinder pressure)
4. Adjust regulator to set delivery pressure:
 - Turn regulator adjustment knob clockwise to increase pressure
 - Set to 0.4–0.5 MPa (58–73 psi) as indicated on low-pressure gauge
 - For SSM-5000A: recommended setting is 0.45 MPa (65 psi)
5. Verify stable pressure reading (no fluctuations)
6. Record cylinder pressure and regulator setting in log book

7.2.2 Step 2: Power On SSM-5000A Main Unit

1. Locate main power switch on rear panel (lower right corner)
2. Switch to ON position (I)
3. Observe front panel display:
 - Display should illuminate within 5 seconds
 - Initial message: "SHIMADZU SSM-5000A SYSTEM CHECK"
 - Self-diagnostic runs for approximately 2–3 minutes
 - Display will show "READY" when complete
4. Listen for internal fan operation (quiet humming sound)
5. If error codes appear (E001, E002, etc.):
 - Record error code
 - Consult troubleshooting section (Section 11)
 - Do not proceed until error is resolved

7.2.3 Step 3: Start Computer and Software

1. Power on dedicated analysis computer
2. Wait for Windows operating system to fully load
3. Locate SSM-5000A Control Software icon on desktop
4. Double-click to launch software
5. Software loading sequence:
 - Splash screen appears (5 seconds)
 - "Searching for instrument..." message displays
 - Connection status window appears
 - Green indicator shows successful connection
 - If red indicator appears: check USB or RS-232 cable connection
6. Main software window opens with tabs: [Setup] [Calibration] [Sample] [Results]
7. Verify software version matches instrument firmware (display on screen)

7.2.4 Step 4: Initialize Furnace System

1. In software, navigate to [Setup] tab
2. Click "Temperature Control" button
3. Set TC furnace temperature:
 - Click "TC Furnace" field
 - Enter: **900°C**
 - Click "Set" button
4. Set TIC furnace temperature:
 - Click "TIC Furnace" field
 - Enter: **200°C**
 - Click "Set" button
5. Click "Start Heat-Up" button
6. Monitor heating progress in status window:
 - Current temperature displays in real-time
 - TC furnace: heats at approximately 10°C/minute (90 minutes to 900°C)
 - TIC furnace: heats at approximately 5°C/minute (40 minutes to 200°C)
7. **IMPORTANT:** Do not leave laboratory during heat-up
8. Verify furnace status lights:
 - Orange = heating
 - Green = at set temperature

7.2.5 Step 5: Initialize NDIR Detector

1. Detector automatically powers on with main unit
2. Navigate to [Setup] > [Detector] in software
3. Verify detector parameters:
 - Detection range: 0–100% CO₂ (auto-range)
 - Response time: Standard (3 seconds)
 - Zero adjustment: Auto
4. Allow 30 minutes minimum for detector stabilization
5. Monitor baseline signal in real-time display window:
 - Initial baseline may drift during warm-up
 - Acceptable drift after stabilization: < 5 ppm/hour
 - Baseline noise: < 2 ppm peak-to-peak
6. Perform automatic zero adjustment:
 - Wait until temperature stability achieved (green indicator)
 - Click "Auto Zero" button
 - System purges detector cell with carrier gas (2 minutes)
 - Baseline resets to zero
 - Verify baseline stability within 5 minutes

7.3 Gas Flow Rate Configuration

1. Navigate to [Setup] > [Gas Control] tab
2. Configure TC analysis gas flows:
 - Carrier gas (O₂): 500 mL/min
 - Purge time before combustion: 30 seconds
 - Combustion duration: 180 seconds (3 minutes)
3. Configure TIC analysis gas flows:
 - Carrier gas (O₂): 300 mL/min
 - Purge time before analysis: 20 seconds
 - Analysis duration: 300 seconds (5 minutes)
4. Verify flow rates using:
 - Software flow indicators (display in mL/min)
 - OR external flow meter at gas outlet (optional verification)
5. Click "Apply Settings" to save configuration

7.4 System Blank Check

1. Purpose: Verify system cleanliness and absence of contamination
2. Pre-combust 5 ceramic boats:
 - Place boats in TC furnace position
 - Heat at 900°C for 10 minutes
 - Cool in desiccator for 30 minutes minimum
3. Load 3 blank boats into autosampler positions 1–3
4. In software, navigate to [Sample] tab
5. Create blank sample sequence:
 - Sample ID: BLANK-01, BLANK-02, BLANK-03
 - Sample Type: Blank
 - Weight: 0.0000 g (enter zero)
 - Analysis: TC only
6. Click "Start Analysis" button
7. System analyzes each blank boat automatically
8. Review results in [Results] tab:
 - Calculate average blank value
 - Acceptable blank: < 0.1 mg C per boat
 - If blank > 0.1 mg C:
 - (a) Re-combust boats at 900°C for 15 minutes
 - (b) Retest blanks
 - (c) If still high: replace boats with new set
 - (d) Check for contamination in combustion tube
9. Record blank values in calibration log book

8 Standard Curve Creation and Verification

8.1 TC Calibration Curve Generation

8.1.1 Prepare Glucose Standard Solutions

Follow Section 5.2 for preparation of 6 concentration levels.

8.1.2 Load Standards into Software

1. Navigate to [Calibration] tab in software
2. Click "New Calibration" button
3. Select calibration type: **TC (Total Carbon)**
4. Click "Standard Table" button
5. Enter standard information in table:

Vial Pos.	Std ID	Conc. (mg C/L)	Inj. Vol. (μ L)	Expected Peak Area	Replicates
1	TC-STD-1	10	100	Low	3
2	TC-STD-2	25	100	Low-Med	3
3	TC-STD-3	50	100	Medium	3
4	TC-STD-4	100	100	Med-High	3
5	TC-STD-5	250	100	High	3
6	TC-STD-6	500	100	Very High	3

Table 4: TC Calibration Standard Entry Table

1. For each standard level:
 - Enter Standard ID in "Sample ID" field
 - Enter concentration in "Concentration (mg C/L)" field
 - Set injection volume to 100 μ L
 - Set number of replicates to 3
2. Click "Save Table" to store standard sequence
3. Load standard vials into autosampler:
 - Position standards in order (low to high concentration)
 - Verify adequate volume in each vial ($> 500 \mu$ L)
 - Check vial caps are properly sealed
 - Ensure autosampler carousel is properly seated

8.1.3 Run TC Calibration Sequence

1. Verify furnaces at target temperature (green status indicators)
2. Verify baseline stability (< 5 ppm drift)
3. Click "Start Calibration" button in software
4. System performs automatic sequence:
 - Injects 100 μ L of standard solution onto pre-combusted boat
 - Transfers boat to TC furnace (900°C)
 - Combusts sample for 3 minutes in oxygen atmosphere

- CO₂ carried to NDIR detector
 - Peak area integrated automatically
 - Repeats for all 3 replicates
 - Advances to next concentration level
5. Monitor analysis progress in real-time:
 - Chromatogram displays each peak
 - Peak area values appear in results table
 - Verify peak shapes are symmetric (no shoulders or tailing)
 - Verify retention times are consistent (within 5 seconds)
 6. Total analysis time: approximately 90 minutes (6 levels $\times$ 3 reps $\times$ 5 min/sample)

8.1.4 Evaluate TC Calibration Curve

1. After sequence completion, click "Calibration Results" button
2. Software displays calibration curve graph:
 - X-axis: Concentration (mg C/L)
 - Y-axis: Peak Area (arbitrary units)
 - Data points shown with error bars (standard deviation)
3. Review regression statistics displayed in results panel:

Parameter	Specification	Action if Failed
Correlation coefficient (r^2)	≥ 0.999	Re-prepare standards
Residuals at each level	$< 10\%$	Investigate outliers
Y-intercept	$< 5\%$ of lowest std response	Check blank contamination
Slope consistency	RSD $< 5\%$ over time	Verify detector stability

Table 5: TC Calibration Acceptance Criteria

1. Calculate residuals for each standard:

$$\text{Residual (\%)} = \frac{|\text{Measured} - \text{Expected}|}{\text{Expected}} \times 100 \quad (2)$$

2. If any standard fails residual criterion:
 - Mark outlier in software (right-click data point > "Exclude")
 - Re-analyze that concentration level only
 - If repeat fails: remake that standard solution
3. If $r^2 < 0.999$:
 - Check for non-linearity (may need quadratic fit)
 - Verify standard preparation calculations
 - Re-prepare entire standard series
 - Check detector linearity with known reference standards
4. When calibration passes all criteria:
 - Click "Accept Calibration" button

- Enter calibration ID: "TC-CAL-YYYYMMDD"
- Software saves calibration curve equation: $y = mx + b$
- Record slope (m), intercept (b), and r^2 in log book
- Print calibration report or save as PDF

8.1.5 Calibration Documentation

Create calibration file containing:

- Calibration curve graph with data points
- Regression equation and statistics
- Table of expected vs. measured values
- Peak chromatograms for all standards
- Analyst name, date, and instrument ID
- Lot numbers of standard materials used

8.2 TIC Calibration Curve Generation

8.2.1 Prepare CaCO₃ Solid Standards

1. Pre-combust ceramic boats at 900°C for 10 minutes
2. Cool in desiccator for 30 minutes
3. Tare analytical balance with empty boat
4. Weigh CaCO₃ directly into boats according to Table 2
5. Record exact weight to nearest 0.1 mg in log book
6. Calculate expected carbon mass:

$$\text{C (mg)} = \text{CaCO}_3 \text{ weight (mg)} \times 0.120 \quad (3)$$

7. Prepare 3 boats at each concentration level (15 boats total)
8. Load boats into autosampler carousel positions 1–15

8.2.2 Load TIC Standards into Software

1. Navigate to [Calibration] tab
2. Click "New Calibration" > **TIC (Total Inorganic Carbon)**
3. Click "Standard Table" button
4. Enter standard information:

Vial Pos.	Std ID	CaCO ₃ (mg)	Carbon (mg)	Acid (μL)	Expected Peak	Reps
1–3	TIC-STD-1	8.3	1.0	150	Low	3
4–6	TIC-STD-2	41.7	5.0	150	Low-Med	3
7–9	TIC-STD-3	83.3	10.0	200	Medium	3
10–12	TIC-STD-4	208.3	25.0	200	Med-High	3
13–15	TIC-STD-5	416.7	50.0	250	High	3

Table 6: TIC Calibration Standard Entry Table

1. For each replicate:
 - Enter Sample ID (e.g., TIC-STD-1-A, TIC-STD-1-B, TIC-STD-1-C)
 - Enter actual CaCO₃ weight measured
 - Enter calculated carbon mass
 - Specify acid volume to be added
2. Click "Save Table"

8.2.3 Run TIC Calibration Sequence

1. Verify TIC furnace temperature: 200°C (green indicator)
2. Verify acid reservoir contains fresh 25% H₃PO₄ (minimum 50 mL)
3. Prime acid delivery system:
 - Click [Setup] > [Acid System] > "Prime"
 - System draws acid through tubing (observe in waste bottle)
 - Prime until no air bubbles visible in tubing (approximately 2 minutes)
 - Click "Stop Prime"
4. Return to [Calibration] tab
5. Click "Start Calibration" button
6. System performs automatic sequence for each standard:
 - (a) Autosampler positions boat at acid addition station
 - (b) Syringe dispenses specified volume of acid onto CaCO₃
 - (c) Reaction begins immediately: $\text{CaCO}_3 + 2\text{H}_3\text{PO}_4 \rightarrow 3\text{CO}_2 + \text{Ca}_3(\text{PO}_4)_2 + 3\text{H}_2\text{O}$
 - (d) Wait 30 seconds for initial vigorous reaction
 - (e) Transfer boat to TIC furnace (200°C)
 - (f) Heat for 5 minutes to ensure complete carbonate decomposition
 - (g) Carrier gas sweeps CO₂ to NDIR detector
 - (h) Peak area integrated automatically
 - (i) Boat moved to waste position
 - (j) System advances to next standard
7. Monitor analysis in real-time:
 - Observe acid delivery (should be smooth, no sputtering)
 - Watch for complete effervescence (bubbling stops within 1–2 minutes)
 - Verify peak shapes are smooth and Gaussian
 - Check for baseline return to zero between samples
8. Total analysis time: approximately 120 minutes (15 samples × 8 min/sample)

8.2.4 Evaluate TIC Calibration Curve

1. Click "Calibration Results" after sequence completion
2. Review calibration curve graph (mg C vs. Peak Area)
3. Examine regression statistics:

Parameter	Specification	Action if Failed
Correlation coefficient (r^2)	≥ 0.998	Re-run calibration
Residuals at each level	$< 15\%$	Check acid addition
Y-intercept	$< 10\%$ of lowest std	Check system blank
Replicate precision (RSD)	$< 10\%$	Improve weighing precision
Recovery at mid-level	90–110%	Verify CaCO ₃ purity

Table 7: TIC Calibration Acceptance Criteria

1. Common issues and troubleshooting:

- **High variability at low levels:** Increase sample size or acid volume
 - **Incomplete reaction:** Extend heating time to 7–10 minutes
 - **Poor peak shape:** Check for acid delivery blockage, clean syringe
 - **Low recovery:** Verify CaCO_3 dried properly, check for moisture uptake
2. When calibration passes all criteria:
 - Click "Accept Calibration"
 - Enter calibration ID: "TIC-CAL-YYYYMMDD"
 - Save calibration file with all documentation
 - Print calibration report for laboratory records

8.3 Calibration Verification with QC Standards

8.3.1 Prepare QC Check Standards

1. **TC QC Standard:** Prepare mid-level glucose standard (100 mg C/L)
 - Use different glucose lot than calibration standards (if available)
 - Prepare fresh from stock solution
 - Target: expected recovery 95–105%
2. **TIC QC Standard:** Weigh 83.3 mg CaCO_3 (10 mg C equivalent)
 - Use different CaCO_3 lot than calibration (if available)
 - Prepare 3 replicates
 - Target: expected recovery 90–110%
3. **Certified Reference Material (CRM):** If available
 - Use certified soil standard with known TC and TIC
 - Examples: NIST SRM 2709a (San Joaquin Soil), NIST SRM 2711a (Montana Soil)
 - Prepare 3 replicates at recommended weight

8.3.2 Analyze QC Standards

1. Navigate to [Sample] tab
2. Create QC sample sequence:

Sample ID	Type	Expected	Accept. Range	Replicates
TC-QC-MID	TC Check	100 mg C/L	95–105 mg C/L	3
TIC-QC-MID	TIC Check	10.0 mg C	9.0–11.0 mg C	3
CRM-2709a	Reference Soil	Per certificate	Per certificate	3

Table 8: QC Standard Verification Sequence

1. Load QC samples into autosampler
2. Click "Start Analysis"
3. System analyzes using active calibration curves
4. Review results and calculate recovery:

$$\text{Recovery (\%)} = \frac{\text{Measured Value}}{\text{Expected Value}} \times 100 \quad (4)$$

5. QC acceptance criteria:

- TC QC recovery: 95–105%
- TIC QC recovery: 90–110%
- CRM recovery: within certified uncertainty range
- Replicate RSD: < 10%

6. If QC fails:

- Re-analyze QC to rule out injection/analysis error
- If repeat fails: investigate calibration
- Possible causes: standard preparation error, detector drift, contamination
- Action: Recalibrate system and re-analyze QC
- Do NOT proceed to sample analysis until QC passes

7. Record QC results in Quality Control Log:

- Date and time
- QC ID and lot number
- Expected and measured values
- Recovery percentage
- Analyst initials
- Pass/Fail designation

9 Sample Table Preparation and Analysis

9.1 Create Sample Analysis Sequence

9.1.1 Sample Organization Strategy

1. Organize samples by expected carbonate content (high, medium, low)
2. Analyze samples with similar carbonate levels together
3. Include appropriate QC samples throughout sequence
4. Recommended sequence structure:

Position	Sample ID	Type	Analysis	Weight	Reps
1	BLANK-01	Method Blank	TC	0.0000 g	1
2–4	TC-QC-MID	QC Standard	TC	100 mg/L	3
5–7	SOIL-001-A	Unknown	TC	50.0 mg	3
8–10	SOIL-001-B	Unknown	TC	50.0 mg	3
11–13	SOIL-002-A	Unknown	TC	45.2 mg	3
14–16	SOIL-002-B	Unknown	TC	45.2 mg	3
17–19	SOIL-003-A	Unknown	TC	52.8 mg	3
20–22	SOIL-003-B	Unknown	TC	52.8 mg	3
23	BLANK-02	Method Blank	TC	0.0000 g	1
24–26	TC-QC-MID	QC Standard	TC	100 mg/L	3
27–29	SOIL-004-A	Unknown	TC	38.5 mg	3
... continue sample sequence ...					

9.1.2 Enter Sample Information into Software

1. Navigate to [Sample] tab in software
2. Click "New Sequence" button
3. Enter sequence information:
 - Sequence Name: "PROJECT-YYYYMMDD-TC"
 - Analyst Name: [Your name]
 - Project ID: [If applicable]
 - Analysis Type: TC or TIC
4. Click "Sample Table" button
5. For each sample position, enter:
 - Vial Position: 1–40 (autosampler position)
 - Sample ID: Unique identifier
 - Sample Type: Unknown/QC/Blank/Duplicate
 - Sample Weight: Exact weight in mg (dry weight basis)
 - Dilution Factor: Usually 1 (adjust if sample pre-diluted)
 - Number of Replicates: 3 (or 5 for low-level samples)
 - Comments: Sample description, carbonate level, special notes
6. Use software features:
 - Copy/Paste function for duplicate entries
 - "Fill Down" to replicate information
 - Import from Excel spreadsheet (if available)
7. Review entire table for errors before saving
8. Click "Save Sequence"
9. File naming convention: "PROJ-ANALYST-YYYYMMDD-TCOR TIC.seq"

9.2 Sample Weighing and Loading

9.2.1 Prepare Sample Boats

1. Pre-combust ceramic boats:
 - Load 40 boats (full carousel) into TC furnace
 - Heat at 900°C for 10 minutes
 - Transfer immediately to desiccator using heat-resistant tongs
 - Cool for minimum 30 minutes
2. Label boats with sample IDs using:
 - High-temperature ceramic marker, OR
 - Small adhesive labels (note: will burn off during analysis)
 - Match boat labels to autosampler positions

9.2.2 Weigh Soil Samples

1. Set up analytical balance in draft-free area
2. Place anti-static mat on balance pan
3. Calibrate balance with certified weights if not auto-calibrating
4. For each sample:

- (a) Close balance doors
 - (b) Press "Tare" button with empty boat on pan
 - (c) Verify zero reading (0.0000 g)
 - (d) Open door slowly to prevent air currents
 - (e) Add soil sample using micro-spatula:
 - Add sample in small increments
 - Target weight according to expected carbonate content
 - High carbonate: 10–20 mg
 - Medium carbonate: 20–40 mg
 - Low carbonate: 40–80 mg
 - (f) Close balance door and wait for stable reading
 - (g) Record weight when stability indicator appears (< 5 seconds)
 - (h) Record weight to 0.0001 g precision in laboratory notebook
 - (i) Transfer boat immediately to labeled position in autosampler
5. Quality checks during weighing:
- Verify boat placement in center of pan
 - Check for powder on boat exterior (clean if necessary)
 - Ensure sample is evenly distributed in boat
 - Avoid touching boat with bare hands (use powder-free gloves or tweezers)
6. Cross-reference:
- Check sample ID on boat matches sequence table
 - Verify autosampler position corresponds to sequence
 - Confirm weight is within acceptable range for carbonate level

9.2.3 Load Autosampler Carousel

1. Remove carousel from instrument:
 - Open autosampler door
 - Release carousel lock mechanism
 - Lift carousel straight up and out
 - Place on clean, level surface
2. Load boats into numbered positions:
 - Match boat to position number in sequence table
 - Ensure boats are fully seated in holders
 - Check boat clips engage properly
 - Verify no boats are tilted or unstable
3. Visual inspection:
 - Count total boats loaded (should match sequence length)
 - Verify empty positions correspond to sequence gaps
 - Check for any damaged or cracked boats
4. Reinstall carousel:
 - Align carousel orientation marks

- Lower carousel into instrument
 - Engage lock mechanism (hear/feel click)
 - Manually rotate carousel to verify smooth operation
 - Close autosampler door
5. Software verification:
- In software, click "Check Carousel"
 - System performs position check (rotates to each position)
 - Verify "Carousel OK" message appears
 - If error: reload carousel and recheck

9.3 TC Analysis Execution

9.3.1 Pre-Analysis Checklist

Verify all items before starting:

TC furnace at 900°C (green indicator)
Baseline stable (< 5 ppm drift)
Oxygen pressure 0.45 MPa (65 psi)
Calibration current and verified
QC standards passed acceptance criteria
Sample sequence saved
All samples weighed and loaded correctly
Carousel installed and operational
Waste containers not full
Computer hard drive has adequate space (> 1 GB)

9.3.2 Start Analysis Sequence

1. In [Sample] tab, select saved sequence file
2. Review sequence summary:
 - Total number of samples
 - Estimated run time
 - Calibration curve to be used
3. Click "Start Analysis" button
4. Confirmation dialog appears:
 - Verify sequence name correct
 - Check starting position (usually Position 1)
 - Confirm analysis parameters
 - Click "OK" to proceed
5. System begins automated analysis:
 - (a) Carousel rotates to first sample position
 - (b) Boat transferred to loading zone
 - (c) Boat moved into TC furnace (900°C)
 - (d) Oxygen flow purges system (30 seconds)
 - (e) Combustion period begins (180 seconds):

- All carbon oxidized to CO₂
 - CO₂ swept to NDIR detector by carrier gas
 - Real-time chromatogram displays on screen
- (f) Peak detection and integration:
- Software identifies peak start/end points
 - Integrates peak area
 - Calculates carbon concentration using calibration curve
- (g) Post-combustion purge (30 seconds)
- (h) Boat moved to cooling zone
- (i) System advances to next sample automatically

9.3.3 Monitor Analysis Progress

1. Real-time monitoring displays:
 - Current sample ID and position
 - Live chromatogram with peak development
 - Current sample number / total samples
 - Estimated time remaining
 - Last 5 results with statistics
2. Quality indicators to watch:
 - Peak shapes: should be smooth, Gaussian, symmetric
 - Retention times: should be consistent (± 5 seconds)
 - Baseline stability: should return to zero between peaks
 - Replicate agreement: RSD should be $< 10\%$
3. Alert conditions:
 - Software displays warning messages for:
 - (a) Peak area outside calibration range
 - (b) Poor peak shape (excessive tailing)
 - (c) High RSD between replicates ($> 15\%$)
 - (d) Baseline drift exceeds limits
 - Investigate warnings immediately
 - Pause sequence if necessary to resolve issues
4. **Important:** Do not leave instrument unattended
 - Check progress every 30–60 minutes
 - Be present at laboratory during entire run if possible
 - If must leave: ensure another trained analyst monitors

9.4 TIC Analysis Execution

After completing TC analysis, perform TIC analysis on same samples:

9.4.1 Prepare TIC Sample Boats

1. Pre-combust fresh boats at 900°C (as before)
2. Cool in desiccator for 30 minutes
3. Weigh same soil samples into new boats:
 - Use identical sample masses as TC analysis
 - Record weights to 0.0001 g
 - Label boats with sample IDs + "-TIC" designation

9.4.2 Acid Addition

1. Verify 25% H₃PO₄ in acid reservoir (> 50 mL)
2. Prime acid delivery system (if first use of day)
3. Load boats into autosampler carousel
4. Create TIC sample sequence in software:
 - Use same sample IDs with "-TIC" suffix
 - Enter same sample weights
 - Specify acid volume for each sample:
 - (a) High carbonate (vigorous reaction): 150–200 µL
 - (b) Medium carbonate: 150 µL
 - (c) Low carbonate: 100–150 µL

9.4.3 Start TIC Analysis

1. Verify TIC furnace at 200°C (green indicator)
2. Select TIC calibration curve
3. Click "Start Analysis"
4. System performs for each sample:
 - (a) Positions boat at acid addition station
 - (b) Dispenses specified volume of H₃PO₄
 - (c) Waits 30 seconds for initial reaction
 - (d) Transfers boat to TIC furnace (200°C)
 - (e) Heats for 5–7 minutes
 - (f) **Reaction:** Carbonates decompose to CO₂ at low temperature
 - (g) Organic carbon remains unoxidized (requires 900°C)
 - (h) CO₂ swept to detector and quantified
5. Monitor for complete carbonate decomposition:
 - Peak should reach plateau and return to baseline
 - Incomplete reaction shows gradual tail
 - If incomplete: increase heating time to 10 minutes

10 Standard Curve Verification and Quality Control

10.1 Within-Run QC Checks

10.1.1 Continuing Calibration Verification (CCV)

1. Analyze QC check standard every 10 samples:

- Insert QC standard into sequence at regular intervals
 - Use mid-level standard (e.g., 100 mg C/L for TC, 10 mg C for TIC)
 - Prepare QC from separate stock solution (independent verification)
2. Acceptance criteria for CCV:
- Recovery: 90–110% of expected value
 - RSD of triplicates: < 10%
 - If QC fails:
 - (a) Flag affected samples in database
 - (b) Re-analyze QC to confirm failure
 - (c) If repeat fails: stop sequence
 - (d) Investigate cause: detector drift, contamination, standard degradation
 - (e) Recalibrate if necessary
 - (f) Re-analyze all samples since last passing QC
3. Document each QC result:
- Record in QC Control Chart
 - Plot recovery vs. time
 - Look for trends (systematic drift)
 - Calculate running statistics (mean, SD, control limits)

10.1.2 Method Blanks

1. Analyze blank boat every 10 samples:
- Pre-combusted ceramic boat with no sample
 - Analyze using same method as samples
 - Expected result: < 0.1 mg C
2. Blank acceptance:
- Blank < 10% of lowest sample concentration: acceptable
 - Blank > 10% of sample: investigate contamination
 - Possible causes: dirty boats, contaminated gases, carryover
3. Corrective actions if blank fails:
- Re-combust boats at 900°C for 15 minutes
 - Replace quartz wool in combustion tube
 - Check for leaks in gas lines
 - Verify gas purity specifications

10.1.3 Duplicate Samples

1. Analyze 10% of samples as duplicates:
- Prepare duplicate from same homogenized soil
 - Weigh independently
 - Analyze as separate sample
 - Label as "-DUP" in sample ID

2. Calculate Relative Percent Difference (RPD):

$$\text{RPD} = \frac{|\text{Sample} - \text{Duplicate}|}{\left(\frac{\text{Sample} + \text{Duplicate}}{2}\right)} \times 100 \quad (5)$$

3. Acceptance criteria:

- RPD < 15% for samples > 0.5% C
- RPD < 20% for samples < 0.5% C
- If RPD exceeds limit:
 - (a) Check sample homogeneity
 - (b) Re-grind and mix sample
 - (c) Analyze third replicate
 - (d) If variability persists: report as heterogeneous sample

10.2 Calibration Curve Stability Assessment

10.2.1 Daily Calibration Check

1. At start of each analysis day:
 - Analyze 3 calibration levels (low, mid, high)
 - Calculate recovery for each level
 - Compare to original calibration curve
2. Acceptable drift:
 - All levels recover within 95–105%
 - Correlation coefficient remains ≥ 0.999 for TC, ≥ 0.998 for TIC
 - Slope change < 10% from original calibration
3. If drift exceeds limits:
 - Perform full recalibration with all standards
 - Investigate cause: detector aging, gas flow changes, contamination
 - Service instrument if necessary

10.2.2 Calibration Frequency Requirements

- Full calibration (6 levels minimum): Every 7 days or 100 samples
- Calibration verification: Beginning of each analysis day
- CCV check standards: Every 10 samples during run
- Recalibration required if:
 - QC recovery outside 90–110%
 - Detector or furnace maintenance performed
 - Gas cylinder replaced
 - More than 14 days since last calibration

10.2.3 Calibration Curve Documentation

Maintain calibration file containing:

1. Calibration date and analyst name
2. Standard preparation records:
 - Chemical lot numbers
 - Weighing records
 - Dilution calculations
 - Expiration dates
3. Raw data:
 - Individual replicate peak areas
 - Average peak areas per level
 - Standard deviations and RSD
4. Calibration curve graph (print or save as PDF)
5. Regression statistics:
 - Equation: $y = mx + b$
 - Slope (m), intercept (b)
 - Correlation coefficient (r^2)
 - Residuals at each level
6. QC verification results
7. Acceptance/rejection decision with signature
8. File in laboratory records for minimum 5 years

11 Data Recording and Calculations

11.1 Raw Data Collection

Record in bound laboratory notebook or electronic laboratory notebook (ELN):

11.1.1 Sample Information

- Sample ID and description
- Sample collection date and location
- Sample preparation date
- Moisture content (with calculation)
- Sample weight (as-received and dry basis)
- Date and time of analysis

11.1.2 Instrument Information

- Instrument ID: Shimadzu SSM-5000A Serial Number
- Analyst name and initials
- Calibration ID and date
- Calibration verification results
- Gas cylinder lot numbers and pressures
- Reagent lot numbers (phosphoric acid, standards)

11.1.3 Analysis Data

- Raw peak areas for each replicate
- Calculated carbon concentrations (mg C)
- Mean, standard deviation, RSD
- QC check results
- Blank values
- Any deviations from SOP

11.2 Detailed Calculation Procedures

11.2.1 Moisture Content Calculation

$$\text{Moisture (\%)} = \frac{\text{Wet Weight}_{\text{soil}} - \text{Dry Weight}_{\text{soil}}}{\text{Wet Weight}_{\text{soil}}} \times 100 \quad (6)$$

Example:

- Crucible weight: 25.3421 g
- Crucible + wet soil: 28.5632 g
- Wet soil weight: $28.5632 - 25.3421 = 3.2211$ g
- Crucible + dry soil (after 105°C): 28.2156 g
- Dry soil weight: $28.2156 - 25.3421 = 2.8735$ g
- Moisture: $\frac{3.2211 - 2.8735}{3.2211} \times 100 = 10.79\%$

11.2.2 Dry Weight Correction

$$\text{Dry Weight}_{\text{soil}} = \text{Wet Weight}_{\text{soil}} \times \left(1 - \frac{\text{Moisture \%}}{100}\right) \quad (7)$$

Example:

- Sample weighed for analysis: 52.3 mg (as-received moisture content)
- Moisture content: 10.79%
- Dry weight: $52.3 \times (1 - 0.1079) = 52.3 \times 0.8921 = 46.7$ mg

11.2.3 Carbon Concentration from Calibration

Software calculates automatically using calibration equation, but manual check:

$$\text{Carbon Mass (mg)} = \frac{\text{Peak Area} - b}{m} \quad (8)$$

where m = slope, b = intercept from calibration curve

Example TC Analysis:

- Peak area: 125,432 (arbitrary units)
- Calibration slope (m): 2,508.6 AU per mg C
- Calibration intercept (b): 1,245 AU
- Carbon mass: $\frac{125,432 - 1,245}{2,508.6} = 49.5$ mg C

11.2.4 Carbon Percentage Calculation

$$\text{Carbon (\%)} = \frac{\text{Carbon Mass (mg)}}{\text{Sample Dry Weight (mg)}} \times 100 \quad (9)$$

Example:

- TC measured: 49.5 mg C
- Sample dry weight: 46.7 mg
- TC (%): $\frac{49.5}{46.7} \times 100 = 106.0\%$
- **Note:** This result >100% indicates error - investigate!
- Likely cause: incorrect sample weight or moisture calculation

Corrected Example:

- TC measured: 2.35 mg C
- Sample dry weight: 46.7 mg
- TC (%): $\frac{2.35}{46.7} \times 100 = 5.03\%$

11.2.5 TOC Calculation

$$\text{TOC (\%)} = \text{TC (\%)} - \text{TIC (\%)} \quad (10)$$

Example:

- TC: 5.03%
- TIC: 2.18%
- TOC: $5.03 - 2.18 = 2.85\%$

Special Case - Negative TOC:

- If $\text{TIC} > \text{TC}$ by small amount ($< 0.5\%$): measurement uncertainty
- Report TOC as "Below Detection Limit" or "0.0%"
- If $\text{TIC} \gg \text{TC}$: serious error in measurement - reanalyze

11.2.6 Statistical Calculations

Mean of Replicates:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} \quad (11)$$

Standard Deviation:

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}} \quad (12)$$

Relative Standard Deviation (RSD):

$$\text{RSD (\%)} = \frac{s}{\bar{x}} \times 100 \quad (13)$$

Example Calculation:

Three TC replicates: 5.12%, 4.98%, 5.07%

- Mean: $\bar{x} = \frac{5.12+4.98+5.07}{3} = 5.06\%$
- Deviations: 0.06, -0.08, 0.01
- Variance: $\frac{(0.06)^2 + (-0.08)^2 + (0.01)^2}{3-1} = \frac{0.0101}{2} = 0.00505$
- Standard deviation: $s = \sqrt{0.00505} = 0.071\%$
- RSD: $\frac{0.071}{5.06} \times 100 = 1.4\%$ (acceptable, $<10\%$)

11.2.7 Relative Percent Difference (RPD)

For duplicate samples:

$$\text{RPD (\%)} = \frac{|x_1 - x_2|}{\left(\frac{x_1 + x_2}{2}\right)} \times 100 \quad (14)$$

Example:

- Sample: 5.06% TC
- Duplicate: 4.89% TC
- RPD: $\frac{|5.06 - 4.89|}{\frac{5.06 + 4.89}{2}} \times 100 = \frac{0.17}{4.975} \times 100 = 3.4\%$ (acceptable)

11.3 Significant Figures and Rounding

- Sample weights: Record to 0.0001 g (4 decimal places)
- Peak areas: Software reports to nearest integer
- Carbon concentrations: Report to 2 decimal places
- Final TC, TIC, TOC results: Report to 2 decimal places
- Standard deviations: Report to same decimal places as mean
- Percent values: Report to 0.01% (2 decimal places)

Rounding Rules:

- If digit to right is < 5: round down
- If digit to right is > 5: round up
- If digit to right is exactly 5: round to nearest even number
- Carry full precision through calculations, round only final result

11.4 Reporting Results

11.4.1 Standard Report Format

For each sample, report:

Sample ID	TC (%)	TIC (%)	TOC (%)	Moisture (%)
SOIL-001	5.06 ± 0.07	2.18 ± 0.12	2.88 ± 0.14	10.8
SOIL-002	3.42 ± 0.05	0.85 ± 0.08	2.57 ± 0.09	8.3
SOIL-003	1.95 ± 0.03	0.12 ± 0.02	1.83 ± 0.04	5.2

Table 10: Example Results Report (mean ± standard deviation, $n = 3$, dry weight basis)

11.4.2 Data Qualifiers

Use appropriate data qualifiers when necessary:

- **J** = Estimated value (QC slightly outside limits but usable)
- **U** = Below detection limit
- **D** = Diluted sample
- ***** = Duplicate RPD exceeded but values reported
- **ND** = Not detected
- **NA** = Not analyzed (e.g., TIC not performed on non-calcareous soil)

11.4.3 Required Documentation

Include with every report:

- Sample identification and description
- Analysis date(s)
- Method reference (this SOP number)
- Analyst name
- Instrument ID
- Calibration information
- QC results summary
- Moisture content
- Reporting basis (dry weight)
- Any deviations or qualifiers
- Uncertainty estimates or confidence intervals

12 Maintenance and Troubleshooting

12.1 Daily Maintenance

Perform at end of each analysis day:

1. **Check gas pressures and flows:**
 - Record oxygen cylinder pressure in log
 - Verify regulator maintains 0.45 MPa
 - Check for pressure drops indicating leaks
2. **Verify furnace temperatures:**
 - TC furnace: $900^{\circ}\text{C} \pm 5^{\circ}\text{C}$
 - TIC furnace: $200^{\circ}\text{C} \pm 5^{\circ}\text{C}$
 - Record actual temperatures in maintenance log
3. **Inspect sample inlet and combustion tube:**
 - Look for sample residue buildup
 - Check quartz wool color (replace if brown/black)
 - Verify no cracks or damage to combustion tube
4. **Clean ceramic boats:**
 - Remove used boats from waste position
 - Inspect for damage (cracks, chips)
 - Discard damaged boats
 - Pre-combust reusable boats at 900°C (10 min)
 - Store cleaned boats in desiccator
5. **Empty waste containers:**
 - Check acid waste bottle level
 - Empty if $> 50\%$ full
 - Check condensate trap
6. **Check desiccant:**
 - Observe indicating desiccant color

- Blue/orange = active (acceptable)
- Pink/white = saturated (replace immediately)

7. Backup data:

- Save all analysis files to network drive
- Export results to Excel or LIMS
- Create backup copy on external drive

12.2 Weekly Maintenance

1. Clean combustion tubes thoroughly:

- Allow furnaces to cool to $< 100^{\circ}\text{C}$
- Remove combustion tubes carefully (wear heat-resistant gloves)
- Inspect for carbon deposits or white residue (carbonates)
- Clean with appropriate brush and deionized water
- Dry completely in oven at 105°C
- Replace quartz wool packing (fresh, pre-combusted)
- Reinstall tubes with new O-rings if needed

2. Clean NDIR detector cell:

- Follow manufacturer procedure for cell cleaning
- Use methanol and lint-free wipes
- Check detector windows for fogging or deposits
- Allow to dry completely before reassembly
- Perform detector zero adjustment after cleaning

3. Verify calibration stability:

- Analyze 3 calibration levels
- Compare to stored calibration curve
- Document any drift
- Recalibrate if needed

4. Check acid delivery system:

- Inspect syringe for bubbles or blockage
- Clean syringe barrel and needle
- Prime system thoroughly
- Test delivery volume accuracy (dispense onto weighing paper, weigh)
- Adjust volume calibration if necessary

5. Leak test gas lines:

- Apply leak detection solution to all fittings
- Pressurize system
- Look for bubble formation
- Tighten connections as needed

6. Inspect autosampler:

- Clean carousel and boat holders
- Check positioning mechanism alignment
- Lubricate moving parts per manufacturer specifications
- Test carousel rotation (manual and automatic)

12.3 Monthly Maintenance

1. Replace oxygen scrubber cartridge:

- Locate moisture/oxygen scrubber in gas line
- Record installation date of current cartridge
- Replace per manufacturer schedule (typically 30 days)
- Install new cartridge with proper orientation
- Purge gas lines after replacement

2. Inspect all O-rings and seals:

- Check furnace door seals
- Inspect gas line O-rings
- Look for cracks, deformation, or hardening
- Replace worn seals
- Apply vacuum grease to new O-rings

3. Clean or replace furnace tubes:

- Remove tubes and inspect for damage
- Heavy carbon buildup: replace tube
- Light buildup: clean with ceramic brush
- Soak stubborn deposits in dilute nitric acid (10%)
- Rinse thoroughly with deionized water
- Dry at 105°C before reinstallation

4. Perform full system calibration:

- Prepare fresh calibration standards
- Run complete calibration sequence (6 levels)
- Verify system performance meets specifications
- Document calibration in maintenance log

5. Analyze certified reference material:

- Use NIST or equivalent CRM
- Analyze in quintuplicate
- Calculate recovery and precision
- Must be within certified range
- Document in QC control charts

6. Update maintenance log:

- Record all maintenance performed
- Note any parts replaced (with part numbers)
- Document any issues observed
- Sign and date maintenance record

12.4 Common Issues and Solutions

Problem	Possible Cause	Solution
High blank values (> 0.1 mg C)	Contaminated boats or combustion tube	Pre-combust boats at 900°C for 15 min; clean combustion tube; replace quartz wool
Poor peak shape (tailing)	Incomplete combustion or contamination	Increase combustion time to 5 min; raise temperature to 920°C; clean combustion tube
Low TIC recovery	Insufficient acid or incomplete reaction	Increase acid volume to 200–250 μ L; extend heating time to 10 min; use higher acid concentration (30%)
Vigorous reaction with high carbonate soils	Rapid CO ₂ evolution	Reduce sample size to 5–10 mg; add acid in small increments (50 μ L); wait between additions
Baseline drift (> 5 ppm/hour)	Detector contamination or temperature instability	Clean NDIR detector cell; verify furnace temperature stability; check gas flow rates
Inconsistent replicates (RSD > 15%)	Sample heterogeneity or weighing errors	Improve grinding (finer particle size); increase mixing time; check balance calibration
Negative TOC values	TIC > TC measurement error	Reanalyze both TC and TIC; verify calibrations; check sample weights; ensure no sample loss
Peak not detected	Sample too small or detector sensitivity low	Increase sample size; check detector settings; verify calibration range; clean detector
Carryover between samples	Incomplete purging or residue in tube	Extend purge time to 60 sec; increase furnace temperature; clean combustion tube more frequently
Software communication error	Cable disconnection or driver issue	Check USB/RS-232 connection; restart software; reboot computer; reinstall drivers if necessary
Furnace not heating	Heating element failure or power issue	Check power supply; verify temperature controller settings; test heating element continuity; replace if faulty
Acid not dispensing	Syringe blockage or empty reservoir	Refill acid reservoir; clean syringe needle; prime system; check for air bubbles in line
Autosampler malfunction	Mechanical jam or position error	Manually rotate carousel to identify obstruction; check boat placement; clean positioning sensors; recalibrate positions
Oxygen pressure drop	Leak or cylinder empty	Check cylinder pressure; perform leak test on all fittings; tighten connections; replace cylinder if low
Moisture in gas lines	Saturated desiccant or condensation	Replace desiccant in moisture trap; check for water in gas lines; install heated gas lines if necessary

12.5 When to Call for Service

Contact Shimadzu service technician if:

- Furnace temperatures cannot be maintained or controlled
- NDIR detector shows erratic or no response after cleaning
- Persistent baseline drift (> 10 ppm/hour) despite troubleshooting
- Calibration cannot achieve $r^2 > 0.99$ with known good standards
- Autosampler positioning repeatedly fails
- Gas leaks cannot be eliminated by tightening fittings
- Software errors persist after reinstallation
- Any unusual sounds, smells, or visible damage to instrument
- Annual preventive maintenance is due

13 Waste Disposal

13.1 Chemical Waste

- **Phosphoric acid waste:**
 - Collect in designated acid waste container
 - Neutralize with solid sodium bicarbonate (NaHCO_3) before disposal
 - Add bicarbonate slowly until pH 6–8 (test with pH paper)
 - Dispose via institutional chemical waste program
 - Label container: "Neutralized Phosphoric Acid Waste"
- **Standard solutions:**
 - Glucose/sucrose solutions: dilute and dispose to sanitary sewer (if permitted)
 - Calcium carbonate: can be neutralized and disposed as solid waste
 - Store expired standards separately, labeled with disposal date
- **Cleaning solvents:**
 - Methanol, acetone: collect in organic waste container
 - Dispose per institutional hazardous waste procedures

13.2 Solid Waste

- **Ceramic boats:**
 - Reuse after pre-combustion cleaning (typical life: 50–100 uses)
 - Discard damaged boats as non-hazardous solid waste
 - If contaminated with heavy metals: dispose as hazardous waste
- **Analyzed soil samples:**
 - Non-contaminated soils: dispose as non-hazardous solid waste
 - Known contaminated soils: return to project manager or dispose per waste characterization
- **Quartz wool and furnace packing:**
 - Non-hazardous solid waste
 - Discard when discolored or compressed
- **Desiccant:**
 - Indicating silica gel: regenerate by heating at 200°C for 2 hours, or discard as non-hazardous waste
 - Drierite: dispose as non-hazardous solid waste when saturated

13.3 Gas Cylinders

- Return empty oxygen cylinders to supplier
- Do not dispose in regular waste
- Close cylinder valve before returning
- Mark cylinder as "EMPTY" with tag

14 References

1. Shimadzu SSM-5000A Total Organic Carbon Analyzer User Manual, Rev. 3.0, Shimadzu Corporation, 2023
2. ASTM D4129-13: Standard Test Method for Total and Organic Carbon in Water by High Temperature Oxidation and by Coulometric Detection
3. ISO 10694:1995: Soil Quality – Determination of Organic and Total Carbon after Dry Combustion (Elementary Analysis)
4. Nelson, D.W. and Sommers, L.E. (1996) Total Carbon, Organic Carbon, and Organic Matter. In: Methods of Soil Analysis Part 3 – Chemical Methods, SSSA Book Series 5, Soil Science Society of America, Madison, WI, pp. 961–1010
5. USEPA Method 9060A: Total Organic Carbon, SW-846 Test Methods for Evaluating Solid Waste
6. Loeppert, R.H. and Suarez, D.L. (1996) Carbonate and Gypsum. In: Methods of Soil Analysis Part 3 – Chemical Methods, SSSA Book Series 5, pp. 437–474

15 Revision History

Version	Date	Changes
1.0	October 2025	Initial release with detailed startup, gas system, standard curve, and sample table

16 Approval

Role	Name	Signature	Date
Author			
Technical Reviewer			
Quality Assurance Manager			
Laboratory Director			